Lesson notes · for you

# Data first

AI · Level 3 · Lesson 1

**Time**
25 min

**Materials**
A notebook and a pen; A phone with a camera, if you have one (or draw your examples); Things to sort: leaves, fruit or handwriting samples

## Learning goals

- Plan a dataset: what to collect, the labels, how many and how varied.
- Spot wrong labels, too few examples and one kind far outnumbering another.
- Collect data fairly: ask permission and keep personal details out.

## Activity: Data detectives

1. Pick (3 min): Choose something to sort into two labels: ripe or unripe fruit, healthy or sick leaves, round or pointed leaves, neat or messy writing.
2. Plan (5 min): Write a four-line plan: what you will collect, the two labels, how many of each, and how you will make them varied (size, light, place, who made them).
3. Collect (8 min): Draw or gather 12 examples and write a label on each. Label one of them wrongly on purpose, and note which one on a hidden slip.
4. Swap (5 min): Give your examples to a partner. They look for the wrong label, count each label, and list what kinds of examples are missing.
5. Fix (4 min): Together, write three things that would make the dataset better: more of a label, more variety, or a label to correct. Then reveal the hidden mistake.

**Works in a group of three or four, or at home with a family member. Use drawings if there is no camera, and never photograph people without asking.**

## Discussion

- Why can’t a model notice that one of its labels is wrong?
- A dataset has 90 healthy leaves and 10 sick ones. What could go wrong?
- Whose permission would you need to collect voices at school?

## Check yourself

- One thing my dataset would be missing, and how I would collect it fairly.

**Try it this week**
Collect twenty labelled examples of something around you: leaves, fruit at the market, or handwriting (ask the writers first). Count how many you have of each label, then write down what your dataset is missing: other kinds, other light, other places.

Worksheet

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Name _____ Class _____ Date _____

**1** **You want to train a model on your classmates' handwriting. What should you do first?**

- ◯ Ask the teacher, not the classmates
- ◯ Ask each one, and leave names off
- ◯ Use old pages they threw away

**2** **Which of these are problems in a dataset? Choose all that apply.**

- ☐ Only six examples in all
- ☐ Photos taken at different times of day
- ☐ Every label checked by two people
- ☐ 90 photos of one label and 10 of the other
- ☐ A sick leaf labelled "healthy"

**3** **You don't need to check the labels: the model will work out which ones are wrong.**

- ◯ True
- ◯ False

**4** **Complete the sentence.**

A good dataset has correct _____, enough examples of each kind and a _____ mix, and the people in it gave their _____.

**Word bank:** rules · names · varied · labels · permission

✎ **What would you like to teach a model to sort? Which labels would you use, and what would be hardest to collect fairly?**

Check your answers

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1. **Ask each one, and leave names off**

   It is their handwriting, so they decide.

2. **A sick leaf labelled "healthy"; Only six examples in all; 90 photos of one label and 10 of the other**

   A wrong label, too few examples and one label far outnumbering the other all weaken a model.

3. **False**

   A model treats every label as the truth.

4. **A good dataset has correct labels, enough examples of each kind and a varied mix, and the people in it gave their permission.**

   Correct labels, enough of each kind, variety and permission are the four checks of a good dataset.